

**THE GENERIC $\mathbb{Z}/8$ -TORSOR IN CHARACTERISTIC 2 STAYS
TWO-DIMENSIONAL OVER EVERY QUADRATIC EXTENSION:
A COMPLETE PROOF OF $\mathrm{ed}_k^{[2]}(\tau_{\mathrm{gen}}) = 2$**

0. STATEMENT AND OUTLINE

Throughout, k is an algebraically closed field of characteristic 2, $G = \mathbb{Z}/8 = \langle g \rangle$, $L = k(z_0, z_1, z_2)$ with G acting by the Witt translation $z \mapsto z + 1$, $K = L^G = k(s_0, s_1, s_2)$ where $s = (s_0, s_1, s_2) = \wp(z)$, and $\tau_{\mathrm{gen}}: \mathrm{Spec} L \rightarrow \mathrm{Spec} K$ is the generic $\mathbb{Z}/8$ -torsor, of class $s \in W_3(K)/\wp W_3(K) = H^1(K, \mathbb{Z}/8)$. For a G -torsor τ over a field $F \supseteq k$, $\mathrm{ed}_k(\tau)$ is the least transcendence degree over k of a subfield $k \subseteq F_0 \subseteq F$ such that $\tau \cong \tau_0 \otimes_{F_0} F$ for a G -torsor τ_0 over F_0 ([BF, §1], [P, Def. 3.1]), and

$$\mathrm{ed}_k^{[d]}(\tau) := \min\{\mathrm{ed}_k(\tau \otimes_F F') : [F' : F] \leq d\}, \quad d_j(\tau) := \min\{d \geq 1 : \mathrm{ed}_k^{[d]}(\tau) \leq j\}$$

([P, Definition 3.1]). The paper [P] proved $\mathrm{ed}_k^{[2]}(\tau_{\mathrm{gen}}) \in \{1, 2\}$ and $d_1(\tau_{\mathrm{gen}}) \in \{2, 3, 4\}$ [P, Theorem 7.1]. The purpose of this note is to settle the second slot.

Theorem 0.1. *Let k be algebraically closed of characteristic 2, $K = k(s_0, s_1, s_2)$ and τ_{gen} the generic $\mathbb{Z}/8$ -torsor of Witt class $s = (s_0, s_1, s_2)$. Then*

$$\mathrm{ed}_k(\tau_{\mathrm{gen}} \otimes_K K') \geq 2 \quad \text{for every field extension } K'/K \text{ of degree 2,}$$

with equality for $K' = K(v_2)$, $v_2^2 + v_2 = s_2$. Consequently

$$\mathrm{ed}_k^{[2]}(\tau_{\mathrm{gen}}) = 2 \quad \text{and} \quad d_1(\tau_{\mathrm{gen}}) \in \{3, 4\}.$$

Remark 0.2. (i) Since $\mathrm{ed}_k(\tau_{\mathrm{gen}} \otimes K') \leq \mathrm{ed}_k(\tau_{\mathrm{gen}}) \leq 3$, the value $\mathrm{ed}_k(\tau_{\mathrm{gen}} \otimes K')$ lies in $\{2, 3\}$ for every quadratic K' , and it equals 2 for all the extensions listed in [P, Theorem 7.1] and [A, Theorem 11.9]; it equals 2 for every quadratic K' as soon as $\mathrm{ed}_k(\mathbb{Z}/8) = 2$. We do not claim the upper bound 2 for an arbitrary quadratic K' : what is proved here, and what $\mathrm{ed}_k^{[2]}(\tau_{\mathrm{gen}}) = 2$ means, is the lower bound. (ii) The theorem is stated for k algebraically closed because every input from [P] is.

Outline of the proof. Suppose $\mathrm{ed}_k(\tau_{\mathrm{gen}} \otimes K') \leq 1$ for some quadratic K'/K . By purely inseparable invariance and the exclusion of $K(z_0)$, $K' = K(v_f)$ is separable with $f \notin \{0, s_0\} + \wp K$, so $L' = L \otimes_K K'$ is a field. The geometric criterion of [P, Theorem 4.2, Corollary 4.4] turns the hypothesis into a smooth projective curve C over k with faithful G -action and a G -equivariant field embedding $\iota: k(C) \hookrightarrow L'$; the fixed field $k(B)$, $B = C/G$, embeds into K' and is the “descent field” F_0 . We pass to the quotient $C_2 = C/\langle g^4 \rangle$ with its faithful $\mathbb{Z}/4$ -action and to its twist E_2 by the $\mathbb{Z}/4$ -class $(s_0, s_1)_K$; the embedding ι produces a point $Q \in E_2(K')$ whose image in B is the descent point, hence is non-constant. The K -rational divisor D_Q of degree 2 on E_2 attached to Q (“ $Q + \sigma Q$ ”) is the heart of the argument: after the base change $L_2 = k(z_0, z_1, s_2)$ that trivialises the twist, its class is that of a *constant* divisor D_0 on C_2 with $\mathbb{Z}/4$ -invariant class, because rational maps from $\mathbb{A}^3$ to a Jacobian are constant (Step 1). Either $\ell(D_0) = 1$, and then D_Q is supported on constant points, forcing the descent point to be constant (Step 2); or $\ell(D_0) = 2$, and then D_0 moves in a pencil $|D_0|$ that is the pullback of a degree-2 map $u: C_2 \rightarrow \mathbb{P}^1$ on which $\mathbb{Z}/4$ acts through $u \mapsto u + 1$ (the hyperelliptic case, Step 3, or the supersingular genus-1

case, Step 4(i)); the twisted coordinate $\tilde{u} = u + z_0$ is K -rational, D_Q is a $\tilde{u}$ -fibre over $\mathbb{P}^1(K)$, and the descent point $t(Q) = \tilde{u}(Q)^2 + \tilde{u}(Q) + s_0$ lies in K . That puts the descent field inside K , which is impossible by the truncation argument of [A, Theorem 11.6(b)], reproved here. The only remaining case is an ordinary elliptic curve C_2 on which $\mathbb{Z}/4$ acts by translation (Step 4(ii)); there the class $(s_0, s_1)_K \otimes K'$ is the isogeny torsor at a point $b \in B(K')$, and the corestriction identity $\text{cores} \circ \delta_{K'} = \delta_K \circ N$ together with $B(K) = B(k)$ gives $2(s_0, s_1) = 0$ in $H^1(K, \mathbb{Z}/4)$, which is false. Section 7 checks that the cases are exhaustive and derives the corollaries; Section 8 lists what is cited without proof.

1. CONVENTIONS AND IMPORTED FACTS

We use the notation of [P, Sections 2–7] and [A, Section 11]; [P] and [A] are cited with their own numbering. All curves are over k unless said otherwise; a *curve over a field F* is a smooth projective geometrically integral F -scheme of dimension 1. For a divisor D on a curve X over F we write $L(D) = H^0(X, \mathcal{O}_X(D))$ and $\ell(D) = \dim_F L(D)$. Fix

$$G_2 := G/\langle g^4 \rangle \cong \mathbb{Z}/4, \quad K_2 := k(s_0, s_1), \quad L_2 := k(z_0, z_1, s_2) = L^{\langle g^4 \rangle}.$$

The generator g of G and its image in G_2 are both written g .

1.1. Witt vectors. $W_n(A) = A^n$ with the ring structure defined by the ghost components $w_m(x) = \sum_{i \leq m} p^i x_i^{p^{m-i}}$; $F(x) = (x_i^p)$ on $\mathbb{F}_p$ -algebras, $V(x) = (0, x_0, \dots, x_{n-2})$, $\wp = F - \text{id}$, $[a] = (a, 0, \dots, 0)$ [P, §2.2]. We record the formulas we use, at $p = 2$.

- Fact 1.1** (Witt arithmetic at $p = 2$). (a) $(a + b)_0 = a_0 + b_0$, $(a + b)_1 = a_1 + b_1 + a_0 b_0$, and $(a + b)_2 = a_2 + b_2 + a_1 b_1 + a_0 b_0(a_1 + b_1) + a_0^3 b_0 + a_0 b_0^3$ [P, (3)].
 (b) $\wp(a_0, a_1) = (a_0^2 + a_0, a_1^2 + a_1 + a_0^3 + a_0^2)$ and $\wp(a)_2 = a_2^2 + a_2 + C(a_0, a_1)$ with C as in [P, (4)].
 (c) $x + V^r(y) = (x_0, \dots, x_{r-1}, y_0, \dots)$ if $x_r = \dots = x_{n-1} = 0$ [P, (2)]; V is additive and $\wp V = V \wp$.
 (d) $2 \cdot (a_0, a_1) = (0, a_0^2)$ in $W_2(A)$ and $2 \cdot (a_0, a_1, a_2) = (0, a_0^2, a_1^2)$ in $W_3(A)$, for every $\mathbb{F}_2$ -algebra A .
 (e) $(0, c) \in \wp W_2(F)$ iff $c \in \wp F$; $(0, c_0, c_1) \in \wp W_3(F)$ iff $(c_0, c_1) \in \wp W_2(F)$ [P, (5)].

Proof. (a), (b), (c) are [P, §2.2]; for the reader's convenience: $w_1(a + b) = w_1(a) + w_1(b)$ reads $(a_0 + b_0)^2 + 2(a + b)_1 = a_0^2 + 2a_1 + b_0^2 + 2b_1$, so $(a + b)_1 = a_1 + b_1 - a_0 b_0$ as a polynomial identity over $\mathbb{Z}$ (the ghost map is injective on $W_2(\mathbb{Z}[a, b])$), and reduction modulo 2 gives (a); (b) is $F - \text{id}$ with F the componentwise square. (d) By (a) with $a = b$: $(2a)_0 = 2a_0 = 0$ and $(2a)_1 = 2a_1 + a_0^2 = a_0^2$; in W_3 , $(2a)_2 = 2a_2 + a_1^2 + a_0^2 \cdot 2a_1 + 2a_0^4 = a_1^2$. (e) If $(0, c) = \wp(x_0, x_1)$ then $x_0^2 + x_0 = 0$, so $x_0 \in \{0, 1\}$, $x_0^3 + x_0^2 = 0$, and $c = x_1^2 + x_1 \in \wp F$; conversely $(0, \wp x_1) = \wp(0, x_1)$ by (b). For W_3 : if $(0, c_0, c_1) = \wp(x)$ then again $x_0 \in \{0, 1\}$, $\wp[x_0] = 0$, and $x = [x_0] + V(x_1, x_2)$ gives $V(c_0, c_1) = \wp V(x_1, x_2) = V \wp(x_1, x_2)$, so $(c_0, c_1) = \wp(x_1, x_2)$ as V is injective on components; the converse is $\wp V = V \wp$. $\square$

Fact 1.2 (Artin–Schreier–Witt theory). *For every field F of characteristic 2 there is a group isomorphism $H^1(F, \mathbb{Z}/2^n) \cong W_n(F)/\wp W_n(F)$, functorial in F , sending the class of a to the torsor $\text{Spec } F[x]/(\wp x = a)$ with $1 \in \mathbb{Z}/2^n$ acting by $x \mapsto x + 1$. Truncation $W_n \rightarrow W_m$ corresponds to $\mathbb{Z}/2^n \rightarrow \mathbb{Z}/2^m$ and $V^{n-m}: W_m \rightarrow W_n$ to the inclusion $\mathbb{Z}/2^m \hookrightarrow \mathbb{Z}/2^n$; both commute with $\wp$ and with base change. [P, §2.2], [Se, Ch. V].*

Lemma 1.3. *$V: W_2(F)/\wp \rightarrow W_3(F)/\wp$ is injective, i.e. $H^1(F, \mathbb{Z}/4) \rightarrow H^1(F, \mathbb{Z}/8)$ is injective. Moreover $s_0 \notin \wp K$, hence $(0, s_0) \neq 0$ in $W_2(K)/\wp$ and $(0, s_0, s_1) \neq 0$ in $W_3(K)/\wp$; in particular $(s_0, s_1) \neq 0$ in $W_2(K)/\wp$.*

Proof. Injectivity is Fact 1.1(e). If $s_0 = a^2 + a$ with $a \in K = k(s_1, s_2)(s_0)$, then a is integral over $k(s_1, s_2)[s_0]$, which is integrally closed, so a is a polynomial in s_0 of some degree d , and $a^2 + a$ has degree $2d \neq 1$: contradiction. The rest follows from Fact 1.1(e), and $(s_0, s_1) \neq 0$ because its truncation s_0 is nonzero in $K/\wp K$. $\square$

1.2. The generic torsor and its truncations. By [P, §2.3, Proposition 3.9], L/K is Galois with group G , its unique quadratic subextension is $K_1 = L^{\langle g^2 \rangle} = K(z_0)$ (with $z_0^2 + z_0 = s_0$), and $L^{\langle g^4 \rangle} = L_2 = k(z_0, z_1, s_2)$; the group $G_2 = G/\langle g^4 \rangle$ acts on L_2 by the Witt translation $(z_0, z_1) \mapsto (z_0, z_1) + (1, 0)$ of W_2 , and $\wp(z_0, z_1) = (s_0, s_1)$ (truncation of $\wp z = s$). Hence

$$T_2 := \text{Spec } L_2 \longrightarrow \text{Spec } K$$

is the $\mathbb{Z}/4$ -torsor of class $(s_0, s_1) \in W_2(K)/\wp$, written $(s_0, s_1)_K$; it is $\tau_2 \otimes_{K_2} K$ where τ_2 is the generic $\mathbb{Z}/4$ -torsor over $K_2 = k(s_0, s_1)$.

- Fact 1.4.** (a) [P, Lemma 2.1] $\text{ed}_k(\tau_{\text{gen}}) = \text{ed}_k(\mathbb{Z}/8) \leq 3$; [P, Prop. 6.1] $\text{ed}_k(\tau_2) = \text{ed}_k(\mathbb{Z}/4) = 2$.
 (b) [P, Lemma 4.1(i)] *If τ is a G -torsor over an infinite field $F \supseteq k$ and t is transcendental over F , then $\text{ed}_k(\tau \otimes F(t)) = \text{ed}_k(\tau)$. Consequently $\text{ed}_k((s_0, s_1)_K) = \text{ed}_k(\tau_2 \otimes_{K_2} K_2(s_2)) = 2$.*
 (c) [P, Prop. 3.2(c)] $\text{ed}_k^{[d]}(\tau_{\text{gen}}) \geq 1$ for $d < 8$.
 (d) [P, Prop. 3.6] *If F'/F is finite purely inseparable, then $\text{ed}_k(\tau \otimes F') = \text{ed}_k(\tau)$; the profile only sees separable extensions.*
 (e) [P, Prop. 3.9] $\text{ed}_k(\tau_{\text{gen}} \otimes K(z_0)) \geq \min(n-1, 2) = 2$.
 (f) [P, Theorem 7.1] $\text{ed}_k^{[2]}(\tau_{\text{gen}}) \leq 2$, witnessed by $K(v_2)$, $v_2^2 + v_2 = s_2$ (class $(s_0, s_1, 0)$, defined over $k(s_0, s_1)$); $\text{ed}_k^{[4]}(\tau_{\text{gen}}) = 1$; $\text{ed}_k(\tau_{\text{gen}}) \in \{2, 3\}$.
 (g) [P, §3.1], [BF, Lemma 2.1(b)] $\text{ed}_k(\tau \otimes F') \leq \text{ed}_k(\tau)$ for every extension F'/F .

1.3. Twists and the geometric criterion. For a quasi-projective G -variety X over k and a G -torsor $\tau: T \rightarrow \text{Spec } F$, the twist is ${}^\tau X = (T \times_k X)/G$, an F -variety; it commutes with base change in F [P, §2.4], [DR, §3].

Fact 1.5 ([P, Fact 2.2(a)], [DR, Lemma 7.3]). *For every field $F' \supseteq F$, ${}^\tau X(F') = \text{Hom}_G(T \times_F F', X)$, the set of G -equivariant k -morphisms.*

Fact 1.6 ([P, Theorem 4.2, (1) $\Rightarrow$ (3) and last paragraph]). *Let $\tau: T \rightarrow \text{Spec } F$ be a G -torsor and suppose $\text{ed}_k(\tau \otimes F') \leq m$ for a field F'/F . Then there is a quasi-projective generically free primitive G -variety Z over k with $\dim Z \leq m$ and a G -equivariant morphism $T \times_F F' \rightarrow Z^{\text{free}}$, where $Z^{\text{free}} = Z \setminus Z^{g^4}$ is the locus of points with trivial stabiliser. Conversely such a map gives $\text{ed}_k(\tau \otimes F') \leq \dim Z$ [P, Lemma 2.3]. If $T \times_F F'$ is the spectrum of a field, Z may be taken irreducible with faithful G -action.*

Fact 1.7 ([P, Lemma 2.4]). *Let Z be an irreducible G -variety with faithful action, T connected, and $f: T \rightarrow Z$ equivariant. The closure Y of the image point of f is irreducible and G -stable, and the image point is free iff G acts faithfully on Y . If $\dim Z = 1$ and the image point is free, then f is dominant.*

Fact 1.8 ([P, Lemma 2.5(ii)]). $\text{PGL}_2(k)$ has no element of order 4. Every element of order 2 of $\text{PGL}_2(k)$ is conjugate to $u \mapsto u + 1$.

Proof. A 2-element of $\text{PGL}_2(k)$ lifts to $A \in \text{GL}_2(k)$ with $A^{2^m} = \lambda$ scalar; replacing A by A/μ with $\mu^{2^m} = \lambda$ (k is algebraically closed) we may assume $A^{2^m} = 1$. Then $(A-1)^{2^m} = A^{2^m} - 1 = 0$, so $N := A - 1$ is nilpotent, $N^2 = 0$, and $A^2 = 1 + 2N = 1$: no element of order 4. If the image

of A has order 2 then $N \neq 0$, so N is conjugate to $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and A to $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, which acts on $\mathbb{P}^1$ by $u \mapsto u + 1$. $\square$

1.4. Abelian varieties, rational maps, Galois descent.

Fact 1.9 (rigidity). *Let A be an abelian variety over k , and let P be a k -variety that is a torsor under A (a principal homogeneous space).*

- (a) *Every k -morphism $\mathbb{P}_k^1 \rightarrow A$ is constant [Mi, Cor. 3.8]; equivalently [P, Lemma 2.5(iv)].*
- (b) *(Weil) Every rational map from a smooth k -variety to A extends to a morphism [Mi, Thm. 3.1], [BLR, 8.4, Cor. 6].*
- (c) *Every k -morphism $\text{Spec } F \rightarrow P$, where $F = k(t_1, \dots, t_m)$ is purely transcendental over k , factors through a k -point of P .*
- (d) [P, Lemma 4.1(ii)] *If $F \supseteq k$ is a field and $F' = F(t_1, \dots, t_m)$ is purely transcendental over F , then $A(F') = A(F)$; the same holds for an abelian variety over F .*

Proof. (a),(b) are cited. (c) Since k is algebraically closed, $P(k) \neq \emptyset$ and $P \cong A$ as k -varieties (translate a k -point to the origin); so a k -morphism $\text{Spec } F \rightarrow P$ is a rational map $\mathbb{A}_k^m \dashrightarrow A$, which by (b) extends to a morphism $\phi: \mathbb{A}^m \rightarrow A$. For $x \in \mathbb{A}^m(k)$ let $\ell_x \subseteq \mathbb{A}^m$ be the line through 0 and x (a point if $x = 0$); $\phi|_{\ell_x}$ extends to $\mathbb{P}^1 \rightarrow A$ and is constant by (a), so $\phi(x) = \phi(0)$ for all x , and ϕ is constant ($\mathbb{A}^m$ is reduced and k algebraically closed). Hence $\text{Spec } F \rightarrow P$ factors through the k -point $\phi(0)$. (d) The proof in [P] is written for an elliptic curve but uses only that A is proper and that $\mathbb{P}_{\bar{F}}^1 \rightarrow A_{\bar{F}}$ is constant (a); it applies verbatim to any abelian variety over F . $\square$

Fact 1.10 (Galois descent). *Let M/F be a finite Galois extension with group Γ .*

- (a) *For F -schemes S, X : $\text{Hom}_F(S, X) \rightarrow \text{Hom}_M(S_M, X_M)$ is injective with image the morphisms commuting with the natural Γ -actions; in particular $X(M)^\Gamma = X(F)$ [BLR, 6.1, Thm. 6 and 6.2, Example B].*
- (b) *A descent datum on a quasi-projective M -scheme Y relative to M/F , i.e. a Γ -action by automorphisms ϕ_γ of the F -scheme Y covering $\text{Spec}(\gamma)$ on $\text{Spec } M$, is effective: there is an F -scheme X , unique up to unique isomorphism, with an M -isomorphism $\psi: X_M \rightarrow Y$ such that $\psi \circ (\text{id}_X \times \text{Spec } \gamma) = \phi_\gamma \circ \psi$ for all γ [BLR, 6.2, Example B], [SeAG, V.20, Prop. 12]. X is a curve over F iff Y is a curve over M (these properties are local for the fpqc topology). F -morphisms $X \rightarrow X'$ between descended schemes are the M -morphisms $Y \rightarrow Y'$ compatible with the descent data (by (a)). If X is a curve then $F(X) = M(Y)^\Gamma$, where Γ acts on $M(Y) = M(X_M) = F(X) \otimes_F M$ through the ϕ_γ^* .*
- (c) *If X is an F -scheme, X_M carries the descent datum $\text{id}_X \times \text{Spec } \gamma$, and (a) is the special case of (b) for this datum. A Galois-invariant divisor on X_M descends uniquely to a divisor on X (for X a curve: closed points of X correspond to Γ -orbits of closed points of X_M , with $x \otimes_F M = \sum_{y \mapsto x} y$).*

Fact 1.11 (Picard schemes). *Let C be a curve over k of genus g .*

- (a) *The Picard functor of C/k is represented by a smooth k -group scheme $\text{Pic}_{C/k}$, locally of finite type, whose identity component $J = \text{Pic}_{C/k}^0$ is an abelian variety of dimension g (the Jacobian) and whose connected components are the $\text{Pic}_{C/k}^d$, $d \in \mathbb{Z}$, each a torsor under J [BLR, 8.2, Thm. 3; 8.4, Thm. 3], [Kl, Thm. 9.4.8, Prop. 9.5.3, Cor. 9.5.4].*
- (b) *Since $C(k) \neq \emptyset$, for every field $F \supseteq k$ the natural map $\text{Pic}(C_F) \rightarrow \text{Pic}_{C/k}(F)$ is bijective and respects degrees [BLR, 8.1, Prop. 4]. Under it, for $\alpha \in \text{Aut}(C)$ the pullback α_F^* on $\text{Pic}(C_F)$ corresponds to $(\alpha^*)_F$, where α^* is the automorphism of the k -group scheme $\text{Pic}_{C/k}$ induced by functoriality.*

Fact 1.12 (cohomology and base change). *Let X be a curve over F , F'/F a field extension, D a divisor on X .*

- (a) $L(D_{F'}) = L(D) \otimes_F F'$; in particular $\ell(D_{F'}) = \ell(D)$ and $\deg D_{F'} = \deg D$ [Ha, III.9.3].
- (b) $\text{Pic}(X) \rightarrow \text{Pic}(X_{F'})$ is injective.
- (c) Two F -morphisms $X \rightarrow Y$ that agree after base change to F' are equal.
- (d) For a nonconstant $h \in F(X)$ and the associated morphism $h: X \rightarrow \mathbb{P}_{F'}^1$, the fibre divisor $h^*(c)$, $c \in \mathbb{P}^1(F)$, is effective, F -rational, of degree $\deg h = [F(X) : F(h)]$, and $h^*(c)_{F'} = (h_{F'})^*(c)$.

Proof. (a) is flat base change. (b) If $\deg D = 0$ and $D_{F'}$ is principal then $\ell(D) = \ell(D_{F'}) = 1$ by (a); a nonzero $h \in L(D)$ has $\text{div } h + D \geq 0$ of degree 0, so $D = \text{div}(h^{-1})$. (c) $\text{Spec } F' \rightarrow \text{Spec } F$ is faithfully flat, hence an epimorphism of schemes. (d) is standard [Ha, II.6.9, II.6.10]; pullback of Cartier divisors commutes with base change. $\square$

1.5. Curves: linear systems, hyperelliptic curves, elliptic curves.

Fact 1.13 (Riemann–Roch and consequences). *Let C be a curve over k of genus g and D a divisor on C .*

- (a) $\ell(D) - \ell(K_C - D) = \deg D + 1 - g$ [Ha, IV.1.3]. If $g = 1$ and $\deg D > 0$ then $\ell(D) = \deg D$.
- (b) If $g \geq 1$ and $\deg D \geq 1$ then $\ell(D) \leq \deg D$.
- (c) If $g \geq 1$, $\deg D = 2$ and $\ell(D) = 2$, then $|D|$ is base point free and defines a morphism $u: C \rightarrow \mathbb{P}^1$ of degree 2 with $D \sim u^*(a)$ for every $a \in \mathbb{P}^1(k)$; u is separable, and $k(C)/k(u)$ is Galois with group $\{1, \iota\}$ generated by an involution ι of C with $C/\iota \cong \mathbb{P}^1$. Conversely the fibres of any degree-2 morphism $C \rightarrow \mathbb{P}^1$ form such a pencil.
- (d) (uniqueness of the g_2^1) If $g \geq 2$, any two divisors D, D' of degree 2 with $\ell(D) = \ell(D') = 2$ are linearly equivalent.

Proof. (a) is cited; for $g = 1$, $K_C = 0$ and $\ell(-D) = 0$. (b) For a point P , $\ell(D) \leq \ell(D - P) + 1$ (evaluation at P in a local coordinate). By induction on $\deg D$ this gives $\ell(D) \leq \ell(D - (\deg D - 1)P) + \deg D - 1$ with $\deg(D - (\deg D - 1)P) = 1$; and a divisor D_1 of degree 1 with $\ell(D_1) \geq 2$ gives a nonconstant h with $\text{div } h + D_1 \geq 0$, hence a degree-1 map $C \rightarrow \mathbb{P}^1$, i.e. $C \cong \mathbb{P}^1$ and $g = 0$. So $\ell(D_1) \leq 1$ and $\ell(D) \leq \deg D$. (c) If P were a base point of $|D|$ then $\ell(D - P) = \ell(D) = 2$ with $\deg(D - P) = 1$, contradicting (b). So $|D|$ is base point free; with $L(D) = \langle h_0, h_1 \rangle$ the map $u = h_1/h_0: C \rightarrow \mathbb{P}^1$ has $u^*(a) = \text{div}(h_1 - ah_0) + D \sim D$ (and $u^*(\infty) = \text{div}(h_0) + D$), and $\deg u = \deg D = 2$. If $k(C)/k(u)$ were inseparable of degree 2, it would be purely inseparable, so $k(C) = k(u^{1/2})$ and $C \cong \mathbb{P}^1$, contradicting $g \geq 1$; a separable extension of degree 2 is Galois, and the nontrivial element of its Galois group is induced by an involution ι of C with $k(C/\iota) = k(C)^\iota = k(u)$. The converse is clear. (d) Let $u, u': C \rightarrow \mathbb{P}^1$ be the degree-2 maps of (c) for D, D' , and let $\Gamma \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ be the image of (u, u') , an irreducible curve of bidegree (a, b) , and $e := \deg(C \rightarrow \Gamma)$. The projections give $eb = \deg u = 2$ and $ea = \deg u' = 2$. If $e = 2$ then $a = b = 1$ and Γ is the graph of an automorphism α of $\mathbb{P}^1$, so $u' = \alpha \circ u$ and $D' \sim u'^*(a_0) = u^*(\alpha^{-1}a_0) \sim D$. If $e = 1$ then Γ has bidegree $(2, 2)$, arithmetic genus $(2 - 1)(2 - 1) = 1$, and C is its normalisation, so $g \leq 1$: excluded. $\square$

Fact 1.14 (elliptic curves in characteristic 2). *Let E be an elliptic curve over k with origin O .*

- (a) $\text{Aut}(E) = E(k) \rtimes \text{Aut}(E, O)$: every automorphism of the curve E is uniquely $t_P \circ \alpha$ with $P \in E(k)$, t_P the translation, $\alpha \in \text{Aut}(E, O)$, and every $\alpha \in \text{Aut}(E, O)$ is a group automorphism, $\alpha \circ t_P = t_{\alpha P} \circ \alpha$ [Sil, III.4.8].
- (b) E is supersingular iff $j(E) = 0$ iff $E(k)[2] = 0$; if E is ordinary then $E(k)[2^r] \cong \mathbb{Z}/2^r$ for all r [Sil, V.3.1, Exercise 5.7]. If $j(E) \neq 0$ then $\text{Aut}(E, O) = \{\pm 1\}$; if $j(E) = 0$ then

$\text{Aut}(E, O)$ has order 24 with Sylow 2-subgroups isomorphic to the quaternion group Q_8 [Sil, App. A, Prop. 1.2], [P, Lemma 2.5(iii)].

- (c) $\text{End}(E)$ is a domain, torsion-free as a $\mathbb{Z}$ -module; for $\phi \in \text{End}(E)$, $\phi + \hat{\phi} = 1 + \deg \phi - \deg(1 - \phi) \in \mathbb{Z}$ and $\phi^2 - (\phi + \hat{\phi})\phi + \deg \phi = 0$ [Sil, III.4.2, III.6.2, III.6.3]. A nonzero isogeny is surjective on k -points, and $\#\ker(\phi)(k) \leq \deg \phi$ [Sil, II.2.3, III.4.10].
- (d) For a finite subgroup $\Phi \subseteq E(k)$ there is an elliptic curve E' and a separable isogeny $\phi: E \rightarrow E'$ with kernel Φ , unique up to isomorphism; if Φ is cyclic of order 4 generated by P , then ϕ is finite étale of degree 4 and $E' \cong E/\langle t_P \rangle$ [Sil, III.4.12].

1.6. Galois cohomology.

Fact 1.15 (restriction, corestriction, connecting maps). *Let F'/F be a finite separable extension of fields, $\Gamma_F \supseteq \Gamma_{F'}$ the absolute Galois groups (for a separable closure $F^s \supseteq F'$), and M a discrete Γ_F -module.*

- (a) $\text{res}: H^q(F, M) \rightarrow H^q(F', M)$ and $\text{cores}: H^q(F', M) \rightarrow H^q(F, M)$ are morphisms of cohomological δ -functors; in degree 0, cores is the norm $M^{\Gamma_{F'}} \rightarrow M^{\Gamma_F}$, $m \mapsto \sum_{\gamma \in \Gamma_F/\Gamma_{F'}} \gamma m$; and $\text{cores} \circ \text{res} = [F' : F]$ [NSW, Prop. 1.5.2, Cor. 1.5.7], [Se, I.2.4].
- (b) For a finite abelian group A with trivial action, $H^1(F, A)$ is the group of isomorphism classes of A -torsors over F (finite étale F -schemes with a simply transitive A -action) [Se, I.5.2]; res is base change; a torsor that has an F -point is trivial; an A -equivariant F -morphism between A -torsors over F is an isomorphism [P, proof of Lemma 2.3]. For $A = \mathbb{Z}/2^n$ this is the group of Fact 1.2.
- (c) Let $0 \rightarrow A \rightarrow E \xrightarrow{\phi} B \rightarrow 0$ be an exact sequence of commutative group varieties over k with ϕ finite étale and A finite constant. Then $0 \rightarrow A \rightarrow E(F^s) \rightarrow B(F^s) \rightarrow 0$ is an exact sequence of Γ_F -modules, and the connecting homomorphism $\delta_F: B(F) \rightarrow H^1(F, A)$ sends b to the class of the A -torsor $\phi^{-1}(b) = E \times_{B, b} \text{Spec } F$ (with A acting by translation) [Se, I.5.4–5.6]. By (a), $\text{cores}_{F'/F} \circ \delta_{F'} = \delta_F \circ N_{F'/F}$ on $B(F')$, where $N_{F'/F}(b) = \sum_{\gamma \in \text{Gal}(F'/F)} \gamma b \in B(F)$ when F'/F is Galois.

Proof. Only (c) needs comment. $E(F^s) \rightarrow B(F^s)$ is surjective because $\phi^{-1}(b)$ is a finite étale F^s -scheme, hence a finite disjoint union of copies of $\text{Spec } F^s$. The description of $\delta_F(b)$: choose $e \in E(F^s)$ with $\phi(e) = b$; the cocycle $\gamma \mapsto \gamma(e) - e \in A$ is by definition $\delta_F(b)$, and it is the cocycle of the torsor $\phi^{-1}(b)$ with respect to its F^s -point e . The last identity is (a) applied to the sequence of Γ_F -modules, restricted to $\Gamma_{F'}$: cores commutes with δ , and on H^0 it is the norm; the sum over $\Gamma_F/\Gamma_{F'}$ is the sum over $\text{Gal}(F'/F)$ when F'/F is Galois. $\square$

Fact 1.16 ([A, Theorem 11.6]). *The statement is reproved below as Theorem 5.1; the only inputs are Facts 1.2, 1.4(a),(b) and inflation–restriction.*

2. SETUP: A HYPOTHETICAL WITNESS AND THE POINT $Q \in E_2(K')$

2.1. Reduction to a separable witness with $L \otimes_K K'$ a field.

Definition 2.1. A *witness* is a separable quadratic extension K'/K with $\text{ed}_k(\tau_{\text{gen}} \otimes_K K') = 1$.

Lemma 2.2. *If $\text{ed}_k(\tau_{\text{gen}} \otimes_K F') \leq 1$ for some field extension F'/K of degree 2, then $F' = K(v_f)$ with $v_f^2 + v_f = f \in K \setminus \wp K$ is a witness and $f \notin \{0, s_0\} + \wp K$, i.e. $F' \neq K(z_0)$.*

Proof. If F'/K were purely inseparable, Fact 1.4(d),(f) would give $\text{ed}_k(\tau_{\text{gen}} \otimes F') = \text{ed}_k(\tau_{\text{gen}}) \geq 2$. So F'/K is separable of degree 2, hence an Artin–Schreier extension $K(v_f)$, $f \notin \wp K$. By Fact 1.4(c), $\text{ed}_k(\tau_{\text{gen}} \otimes F') \geq 1$, so it equals 1. By Fact 1.4(e), $F' \neq K(z_0)$; since $z_0^2 + z_0 = s_0$, $K(z_0) = K(v_{s_0})$, and two Artin–Schreier extensions $K(v_f)$, $K(v_{f'})$ coincide iff $f \equiv f' \pmod{\wp K}$ (Fact 1.2 for $n = 1$), so $f \not\equiv s_0$; and $f \not\equiv 0$ as $[F' : K] = 2$. $\square$

From now on $K' = K(v_f)$ is a witness, σ denotes the generator of $\text{Gal}(K'/K)$, and $f \notin \{0, s_0\} + \wp K$.

Lemma 2.3. *Put $L' := L \otimes_K K'$ and $L'_2 := L_2 \otimes_K K'$.*

- (a) *L' is a field, Galois over K' with group G (acting through the first factor), and $(L')^G = K'$. The involution σ acts on L' through the second factor and commutes with G .*
- (b) *$L'_2 = (L')^{\langle g^4 \rangle}$ is a field, Galois over K' with group G_2 , and $\text{Spec } L'_2 \rightarrow \text{Spec } K'$ is the base change $T_2 \times_K K'$, i.e. the $\mathbb{Z}/4$ -torsor of class $\text{res}_{K'/K}(s_0, s_1) = (s_0, s_1)_K \otimes K'$.*

Proof. (a) $L' = L[v]/(v^2 + v - f)$, which is a field iff $f \notin \wp L$. By inflation–restriction for L/K (Galois with group G , trivial action on $\mathbb{Z}/2$), $\ker(H^1(K, \mathbb{Z}/2) \rightarrow H^1(L, \mathbb{Z}/2)) = H^1(G, \mathbb{Z}/2) = \text{Hom}(\mathbb{Z}/8, \mathbb{Z}/2) \cong \mathbb{Z}/2$, generated by the class of the unique quadratic subextension $K(z_0)$ of L/K , which is $[s_0]$. Hence $f \in \wp L$ iff $f \in \{0, s_0\} + \wp K$, which is excluded. Thus L' is a field; it is the base change of the Galois extension L/K , so it is Galois over K' with group G and fixed field $K \otimes_K K' = K'$. The statement about σ is clear. (b) $L_2 \otimes_K K' \subseteq L \otimes_K K'$ is a K' -subalgebra of a field, finite over K' , hence a field, of degree $[L_2 : K] = 4$ over K' ; it is fixed by g^4 , and $(L')^{\langle g^4 \rangle}$ has degree $[L' : K']/2 = 4$ over K' by Galois theory. So they coincide. The rest is the definition of res (Fact 1.15(b)) and the description of T_2 in Section 1. $\square$

2.2. The witness curve and the descent field.

Proposition 2.4. *There exist a curve C over k with a faithful action of G and a G -equivariant k -algebra embedding $\iota: k(C) \hookrightarrow L'$.*

Proof. By Fact 1.6 applied to τ_{gen} and $F' = K'$ (with $m = 1$; note $T \times_K K' = \text{Spec } L'$ is the spectrum of a field by Lemma 2.3), there are an irreducible quasi-projective G -variety Z with faithful action, $\dim Z \leq 1$, and a G -equivariant morphism $x: \text{Spec } L' \rightarrow Z^{\text{free}}$. If $\dim Z = 0$ the converse part of Fact 1.6 gives $\text{ed}_k(\tau_{\text{gen}} \otimes K') = 0$, contradicting Definition 2.1; so Z is a curve. By Fact 1.7, x is dominant. Let C be the smooth projective model of Z ; the G -action on Z extends to C (functoriality of normalisation and of the projective closure of a curve), faithfully, and the dominant x corresponds to a k -embedding $\iota: k(C) = k(Z) \hookrightarrow L'$, equivariant because x is. $\square$

Fix C and ι as in Proposition 2.4. Since G acts faithfully on C , it acts faithfully on $k(C)$, and by Artin's theorem $k(C)/k(C)^G$ is Galois with group G . Let $B := C/G$ be the curve with $k(B) = k(C)^G$, and $t: C \rightarrow B$ the quotient morphism (finite of degree 8). As ι is equivariant, $\iota(k(B)) \subseteq (L')^G = K'$.

Definition 2.5. The *descent field* of the witness is $F_0 := \iota(k(B)) \subseteq K'$, a subfield containing k with $\text{trdeg}_k F_0 = 1$. The *descent class* is $\beta \in W_3(F_0)/\wp W_3(F_0) = H^1(F_0, \mathbb{Z}/8)$, the class of the G -torsor $\text{Spec } k(C) \rightarrow \text{Spec } k(B)$ transported along $\iota|_{k(B)}: k(B) \xrightarrow{\sim} F_0$.

Lemma 2.6. *$s \equiv \beta$ in $W_3(K')/\wp W_3(K')$; that is, $\tau_{\text{gen}} \otimes K'$ is the base change to K' of a $\mathbb{Z}/8$ -torsor defined over F_0 .*

Proof. The K' -algebra $k(C) \otimes_{k(B)} K'$ (with $k(B) \rightarrow K'$ the map $\iota|_{k(B)}$) is the base change of the G -torsor $\text{Spec } k(C) \rightarrow \text{Spec } k(B)$ along $\text{Spec } K' \rightarrow \text{Spec } k(B)$, so it is a G -torsor over K' of class $\text{res}_{K'/F_0}(\beta)$. The map $k(C) \otimes_{k(B)} K' \rightarrow L'$, $a \otimes c \mapsto \iota(a)c$, is a well-defined G -equivariant K' -algebra homomorphism between G -torsors over K' , hence an isomorphism (Fact 1.15(b)). Since L'/K' is $\tau_{\text{gen}} \otimes K'$, of class $\text{res}_{K'/K}(s)$, the claim follows from Fact 1.2. $\square$

2.3. The $\mathbb{Z}/4$ -quotient and its twist. Let $C_2 := C/\langle g^4 \rangle$, the curve with $k(C_2) = k(C)^{\langle g^4 \rangle}$.

Lemma 2.7. $G_2 = G/\langle g^4 \rangle$ acts faithfully on C_2 , with $C_2/G_2 = B$, and $\iota_2 := \iota|_{k(C_2)}: k(C_2) \hookrightarrow (L')^{\langle g^4 \rangle} = L'_2$ is a G_2 -equivariant k -embedding. The corresponding morphism $x: \operatorname{Spec} L'_2 \rightarrow C_2$ is dominant and satisfies $x \circ \operatorname{Spec}(g) = g_{C_2} \circ x$ for $g \in G_2$, where $\operatorname{Spec}(g)$ is the automorphism of $\operatorname{Spec} L'_2$ induced by the field automorphism g of L'_2 .

Proof. The elements of G acting trivially on $k(C_2) = k(C)^{\langle g^4 \rangle}$ form the subgroup of $G = \operatorname{Gal}(k(C)/k(B))$ fixing this intermediate field pointwise, which is $\langle g^4 \rangle$ by the fundamental theorem of Galois theory. Hence G_2 acts faithfully on $k(C_2)$, i.e. on C_2 , and $k(C_2)^{G_2} = k(C)^G = k(B)$. The image of ι_2 lies in $(L')^{\langle g^4 \rangle} = L'_2$ (Lemma 2.3(b)) by equivariance. A k -embedding of the function field of a curve into a field is the same as a morphism from the spectrum of that field whose image is the generic point, and the identity $\iota_2 \circ g_{C_2}^* = g \circ \iota_2$ translates into $x \circ \operatorname{Spec}(g) = g_{C_2} \circ x$. $\square$

Construction 2.8 (the twist E_2). Let $Y := C_2 \times_k \operatorname{Spec} L_2$, an L_2 -curve, and for $g \in G_2$ let

$$\phi_g := g_{C_2} \times \operatorname{Spec}(g): Y \rightarrow Y,$$

where $\operatorname{Spec}(g)$ is induced by the field automorphism g of L_2 . Each ϕ_g is an automorphism of the k -scheme Y covering $\operatorname{Spec}(g)$ on $\operatorname{Spec} L_2$, and $g \mapsto \phi_g$ is a group homomorphism because G_2 is abelian ($\operatorname{Spec}(g) \circ \operatorname{Spec}(h) = \operatorname{Spec}(hg) = \operatorname{Spec}(gh)$). So $(\phi_g)_g$ is a descent datum on Y relative to the Galois extension L_2/K (whose group is G_2 , Section 1). By Fact 1.10(b) it is effective: there is a curve E_2 over K with an L_2 -isomorphism

$$\psi: E_2 \otimes_K L_2 \xrightarrow{\sim} Y = C_2 \otimes_k L_2, \quad \psi \circ (\operatorname{id}_{E_2} \times \operatorname{Spec} g) = \phi_g \circ \psi \quad (g \in G_2).$$

E_2 is the twist $^{(s_0, s_1)K} C_2 = (T_2 \times_k C_2)/G_2$ of [P, §2.4] (the quotient of $Y = T_2 \times_k C_2$ by the free action ϕ), but we shall only use the following consequences of the descent description, which agree with Fact 1.5.

- Lemma 2.9.** (a) For every field $F \supseteq K$ the map $P \mapsto x_P := \operatorname{pr}_1 \circ \psi \circ (P \times_K \operatorname{id}_{\operatorname{Spec} L_2})$ is a bijection from $E_2(F)$ onto the set of k -morphisms $x: \operatorname{Spec}(F \otimes_K L_2) \rightarrow C_2$ with $x \circ \operatorname{Spec}(\operatorname{id}_F \otimes g) = g_{C_2} \circ x$ for all $g \in G_2$.
- (b) $K(E_2) = L_2(Y)^{G_2}$, where $L_2(Y) = \operatorname{Frac}(k(C_2) \otimes_k L_2)$ and g acts through ϕ_g^* , i.e. by $g_{C_2}^* \otimes g$ on $k(C_2) \otimes_k L_2$. Every element h of $L_2(Y)^{G_2}$, viewed in $K(E_2)$, defines a K -morphism $h: E_2 \rightarrow \mathbb{P}_K^1$ whose base change to L_2 is (via ψ) the morphism $Y \rightarrow \mathbb{P}_{L_2}^1$ defined by h .
- (c) There is a unique K -morphism $t_E: E_2 \rightarrow B_K$ with $(t \times \operatorname{id}) \circ \psi = t_E \otimes \operatorname{id}_{L_2}$. For $P \in E_2(F)$, $t_E(P) \in B_K(F) = B(F)$ is the unique F -point b of B with $b \circ \operatorname{pr} = t \circ x_P$, where $\operatorname{pr}: \operatorname{Spec}(F \otimes_K L_2) \rightarrow \operatorname{Spec} F$.

Proof. (a) By Fact 1.10(a),(b), $\operatorname{Hom}_K(\operatorname{Spec} F, E_2)$ is the set of L_2 -morphisms $y: \operatorname{Spec}(F \otimes_K L_2) \rightarrow Y$ with $\phi_g \circ y = y \circ \operatorname{Spec}(\operatorname{id}_F \otimes g)$ for all g , the bijection being $P \mapsto \psi \circ (P \times \operatorname{id})$. An L_2 -morphism y to $Y = C_2 \times_k \operatorname{Spec} L_2$ is the same as its first component $x = \operatorname{pr}_1 \circ y$, a k -morphism to C_2 (the second component is the structure map). The compatibility condition has first component $g_{C_2} \circ x = x \circ \operatorname{Spec}(\operatorname{id} \otimes g)$ and second component $\operatorname{Spec}(g) \circ \operatorname{pr}_2 \circ y = \operatorname{pr}_2 \circ y \circ \operatorname{Spec}(\operatorname{id} \otimes g)$, which holds automatically. (b) is Fact 1.10(b); a rational function on a curve over a field defines a morphism to $\mathbb{P}^1$ compatibly with base change. (c) $t \times \operatorname{id}: Y \rightarrow B \times_k \operatorname{Spec} L_2 = B_K \otimes_K L_2$ satisfies $(\operatorname{id}_B \times \operatorname{Spec} g) \circ (t \times \operatorname{id}) = (t \times \operatorname{id}) \circ \phi_g$ because $t \circ g_{C_2} = t$; by Fact 1.10(b) it descends to a unique t_E . For $P \in E_2(F)$, $(t_E(P)) \times \operatorname{id}_{L_2} = (t \times \operatorname{id}) \circ \psi \circ (P \times \operatorname{id})$ has first component $t \circ x_P$, and the first component of $b \times \operatorname{id}_{L_2}$ is $b \circ \operatorname{pr}$; finally $b \mapsto b \circ \operatorname{pr}$ is injective because pr is faithfully flat, hence an epimorphism. $\square$

Definition 2.10. $Q \in E_2(K')$ is the point with $x_Q = x$, the dominant equivariant morphism of Lemma 2.7 (note $K' \otimes_K L_2 = L'_2$, and $\text{Spec}(\text{id}_{K'} \otimes g)$ is the $\text{Spec}(g)$ of Lemma 2.7).

Lemma 2.11 (the descent point; Q is non-constant). *Let $b := t_E(Q) \in B(K')$. Then the morphism $b: \text{Spec } K' \rightarrow B$ has image the generic point of B , and the induced map $b^*: k(B) \rightarrow K'$ equals $\iota|_{k(B)}$; hence $b^*(k(B)) = F_0$. In particular b does not factor through any k -point of B .*

Proof. By Lemma 2.9(c), $b \circ \text{pr} = t \circ x$. The morphism $t \circ x$ has image the generic point of B (x is dominant, t surjective) and induces $\iota \circ t^* = \iota|_{k(B)}$ on $k(B)$; hence b has image the generic point, and $\text{pr}^* \circ b^* = \iota|_{k(B)}$ with pr^* the inclusion $K' \subseteq L'_2$. $\square$

Definition 2.12 (the involution and the divisor D_Q). $\text{Gal}(K'/K) = \{1, \sigma\}$ acts on $E_2(K')$ by $P \mapsto \sigma P := P \circ \text{Spec}(\sigma)$. Let $\xi_Q \in E_2$ be the image of Q , a closed point whose residue field $\kappa(\xi_Q)$ is embedded in K' by Q , so $\deg \xi_Q := [\kappa(\xi_Q) : K] \in \{1, 2\}$. Put

$$D_Q := \frac{2}{\deg \xi_Q} \xi_Q,$$

an effective K -rational divisor of degree 2 on E_2 , and $\Delta := \psi(D_Q \otimes_K L_2)$, an effective divisor of degree 2 on $Y = C_2 \otimes_k L_2$.

Lemma 2.13. (a) $D_Q \otimes_K K' = Q + \sigma Q$ as divisors on $E_2 \otimes_K K'$.

(b) $\phi_g(\Delta) = \Delta$ for all $g \in G_2$.

(c) $\text{Supp}(D_Q \otimes_K L_2) = \xi_Q \times_K \text{Spec } L_2$, and the morphism $Q \times_K \text{id}_{L_2}: \text{Spec } L'_2 \rightarrow E_2 \otimes_K L_2$ factors through this closed subscheme; its composite with $\text{pr}_1 \circ \psi$ is x .

Proof. (a) If $\deg \xi_Q = 2$ then $\kappa(\xi_Q) \cong K'$ and $\xi_Q \otimes_K K' = \text{Spec}(K' \otimes_K K')$ consists of the two K' -points given by the two K -embeddings $\kappa(\xi_Q) \rightarrow K'$, namely Q and σQ , which are distinct; so $D_Q \otimes K' = \xi_Q \otimes K' = Q + \sigma Q$. If $\deg \xi_Q = 1$ then $\xi_Q \otimes K'$ is the single K' -point $Q = \sigma Q$ and $D_Q \otimes K' = 2\xi_Q \otimes K' = Q + \sigma Q$. (b) $D_Q \otimes_K L_2$ is invariant under $\text{id} \times \text{Spec } g$, and ψ intertwines $\text{id} \times \text{Spec } g$ with ϕ_g . (c) Q factors through ξ_Q , so $Q \times \text{id}$ factors through $\xi_Q \times_K \text{Spec } L_2$, which is the support of $D_Q \otimes L_2$; the last assertion is Definition 2.10. $\square$

3. STEP 1: THE CLASS OF D_Q IS CONSTANT

Proposition 3.1. *There is an effective divisor $D_0 = P_1 + P_2$ on C_2 , with $P_1, P_2 \in C_2(k)$, such that*

(a) $\Delta \sim D_0 \otimes_k L_2$ on $Y = C_2 \otimes_k L_2$;

(b) $g^* D_0 \sim D_0$ on C_2 for every $g \in G_2$.

Proof. (i) *The class of Δ is constant.* By Fact 1.11(b) (with $F = L_2$; $C_2(k) \neq \emptyset$ as k is algebraically closed) the class $[\Delta] \in \text{Pic}(Y)$ is an L_2 -point $\lambda: \text{Spec } L_2 \rightarrow \text{Pic}_{C_2/k}$, of degree 2, i.e. a k -morphism $\text{Spec } L_2 \rightarrow \text{Pic}_{C_2/k}^2$. By Fact 1.11(a), $\text{Pic}_{C_2/k}^2$ is a torsor under the Jacobian J of C_2 . The field $L_2 = k(z_0, z_1, s_2)$ is purely transcendental of transcendence degree 3 over k : $K = k(s_0, s_1, s_2) \subseteq L_2 \subseteq L$ with $[L : K] = 8$, so $\text{trdeg}_k L_2 = 3$ and the three generators are algebraically independent. Hence, by Fact 1.9(c), λ factors through a k -point $\lambda_0 \in \text{Pic}_{C_2/k}^2(k) = \text{Pic}^2(C_2)$ (Fact 1.11(b) with $F = k$), say $\lambda_0 = [D_1]$ with D_1 a divisor of degree 2 on C_2 . The identification of Fact 1.11(b) is natural in F , so the composite $\text{Spec } L_2 \rightarrow \text{Spec } k \xrightarrow{\lambda_0} \text{Pic}_{C_2/k}^2$ is the class of $D_1 \otimes_k L_2$. Thus $\Delta \sim D_1 \otimes_k L_2$ on Y .

(ii) *The class contains an effective divisor.* Since Δ is effective, $1 \in L(\Delta)$, so $\ell(\Delta) \geq 1$; linear equivalence preserves ℓ , and $\ell(D_1 \otimes L_2) = \ell(D_1)$ by Fact 1.12(a). So $\ell(D_1) \geq 1$: choose $0 \neq h \in L(D_1)$ and put $D_0 := D_1 + \text{div } h$, an effective divisor of degree 2 on C_2 , linearly equivalent

to D_1 . Since k is algebraically closed, $D_0 = P_1 + P_2$ with $P_i \in C_2(k)$. As $\text{div}(h) \otimes L_2 = \text{div}(h)$, $D_0 \otimes L_2 \sim D_1 \otimes L_2 \sim \Delta$, which is (a).

(iii) *Invariance.* Let $g \in G_2$. Writing $D_0 \otimes L_2 = \text{pr}_1^* D_0$ we get

$$\phi_g^*(D_0 \otimes L_2) = (\text{pr}_1 \circ \phi_g)^* D_0 = (g_{C_2} \circ \text{pr}_1)^* D_0 = \text{pr}_1^*(g_{C_2}^* D_0) = (g^* D_0) \otimes L_2.$$

By Lemma 2.13(b), $\phi_g^* \Delta = \Delta$ (apply (b) to g^{-1}). Pullback along the automorphism ϕ_g preserves linear equivalence, so (a) gives $\Delta = \phi_g^* \Delta \sim \phi_g^*(D_0 \otimes L_2) = (g^* D_0) \otimes L_2$, hence $(g^* D_0) \otimes L_2 \sim D_0 \otimes L_2$ on Y . By Fact 1.12(b) ($\text{Pic}(C_2) \rightarrow \text{Pic}(Y)$ is injective), $g^* D_0 \sim D_0$ on C_2 . $\square$

Remark 3.2. The referee's formulation " $\text{Pic}_{E_2/K}$ is the twist of $\text{Pic}_{C_2/k}$ " is not needed: we only use $\text{Pic}(Y) = \text{Pic}_{C_2/k}(L_2)$ for the constant curve C_2 , where the Picard scheme is over k and C_2 has k -points. The step " $\ell(D_1) \geq 1$, so D_0 may be taken effective" is the first of the two gaps flagged in the referee report; without it, $\ell(D_0) \in \{1, 2\}$ below would be unjustified (a general class of degree 2 on a curve of genus ≥ 3 has $\ell = 0$).

4. STEP 2: THE DICHOTOMY $\ell(D_0) \in \{1, 2\}$, AND $\ell(D_0) = 1$ IS IMPOSSIBLE

Lemma 4.1. $g(C_2) \geq 1$.

Proof. If $g(C_2) = 0$ then $C_2 \cong \mathbb{P}_k^1$ and the faithful action of $G_2 \cong \mathbb{Z}/4$ embeds $\mathbb{Z}/4$ into $\text{Aut}(\mathbb{P}^1) = \text{PGL}_2(k)$, contradicting Fact 1.8. $\square$

Corollary 4.2. $\ell(D_0) \in \{1, 2\}$.

Proof. D_0 is effective, so $\ell(D_0) \geq 1$; and $\ell(D_0) \leq \deg D_0 = 2$ by Fact 1.13(b) and Lemma 4.1. $\square$

Proposition 4.3 ($\ell(D_0) = 1$ is impossible). $\ell(D_0) = 2$.

Proof. Suppose $\ell(D_0) = 1$. By Fact 1.12(a), $\ell(D_0 \otimes L_2) = 1$, and by Proposition 3.1(a), $\Delta = D_0 \otimes L_2 + \text{div } h$ with $0 \neq h \in L(D_0 \otimes L_2) = L_2 \cdot 1$; so h is a nonzero constant and

$$\Delta = D_0 \otimes_k L_2 = (P_1 \times \text{Spec } L_2) + (P_2 \times \text{Spec } L_2)$$

as divisors on Y , where $P_i \times \text{Spec } L_2$ are the L_2 -rational points of Y over P_i . In particular $\text{Supp } \Delta \subseteq \{P_1 \times \text{Spec } L_2, P_2 \times \text{Spec } L_2\}$. By Lemma 2.13(c), the morphism $\psi \circ (Q \times_K \text{id}_{L_2}) : \text{Spec } L_2' \rightarrow Y$ factors through $\psi(\text{Supp}(D_Q \otimes L_2)) = \text{Supp } \Delta$. Since $\text{Spec } L_2'$ is a one-point space, its image is one of the points $P_i \times \text{Spec } L_2$, and composing with $\text{pr}_1 : Y \rightarrow C_2$ we find that $x = \text{pr}_1 \circ \psi \circ (Q \times \text{id})$ has image the k -point $P_i \in C_2$. This contradicts the dominance of x (Lemma 2.7). Hence $\ell(D_0) \neq 1$, and Corollary 4.2 gives $\ell(D_0) = 2$. $\square$

Remark 4.4. This covers in particular every non-hyperelliptic C_2 of genus ≥ 2 (for which $\ell(D) \leq 1$ for every divisor of degree 2, by Fact 1.13(c)), and every hyperelliptic C_2 of genus ≥ 2 with $D_0 \not\sim g_2^1$. The argument also covers the case $Q \in E_2(K)$, where $D_Q = 2\xi_Q$. Note that the contradiction is obtained from the dominance of x , i.e. from the non-constancy of the descent point (Lemma 2.11): a constant x would give a constant $b = t_E(Q)$.

Corollary 4.5. *There is a morphism $u : C_2 \rightarrow \mathbb{P}_k^1$ of degree 2 such that $D_0 \sim u^*(a)$ for every $a \in \mathbb{P}^1(k)$; $k(C_2)/k(u)$ is Galois of degree 2, generated by an involution ι_u of C_2 with $C_2/\iota_u \cong \mathbb{P}^1$. If $g(C_2) \geq 2$, then every degree-2 morphism $u' : C_2 \rightarrow \mathbb{P}^1$ satisfies $u'^*(a) \sim D_0$, and $u' = \alpha \circ u$ for an automorphism α of $\mathbb{P}^1$.*

Proof. Fact 1.13(c),(d) applied to D_0 (Proposition 4.3, Lemma 4.1); for the last claim, $u^*(a)$ has $\ell = 2$ and degree 2 (Fact 1.13(c), converse part), so $u'^*(a) \sim D_0 \sim u^*(a')$, and the proof of Fact 1.13(d) (case $e = 2$) gives $u' = \alpha \circ u$. $\square$

5. STEP 3: THE HYPERELLIPTIC CASE, VIA A TWISTED PENCIL

5.1. No descent field inside K .

Theorem 5.1 ([A, Theorem 11.6(b)]). *Let $K' = K(v_f)$ be a separable quadratic extension of K , $F_0 \subseteq K'$ a subfield containing k with $\text{trdeg}_k F_0 = 1$, and $\beta \in W_3(F_0)$ with $s \equiv \beta \pmod{\wp W_3(K')}$. Then $F_0 \not\subseteq K$.*

Proof. Suppose $F_0 \subseteq K$. Then s and β both lie in $W_3(K)$ and have the same image in $H^1(K', \mathbb{Z}/8) = W_3(K')/\wp$. Inflation–restriction for K'/K , with trivial action on $\mathbb{Z}/8$, gives the exact sequence

$$0 \rightarrow H^1(\text{Gal}(K'/K), \mathbb{Z}/8) \rightarrow H^1(K, \mathbb{Z}/8) \rightarrow H^1(K', \mathbb{Z}/8),$$

with $H^1(\text{Gal}(K'/K), \mathbb{Z}/8) = \text{Hom}(\mathbb{Z}/2, \mathbb{Z}/8) \cong \mathbb{Z}/2$, whose nonzero element is the image of the class $[f] \in H^1(K, \mathbb{Z}/2)$ of K'/K under the map induced by the inclusion $\mathbb{Z}/2 \hookrightarrow \mathbb{Z}/8$, i.e. (Fact 1.2) $V^2[f] = (0, 0, f)$. Hence

$$s \equiv \beta + \varepsilon (0, 0, f) \pmod{\wp W_3(K)}, \quad \varepsilon \in \{0, 1\}.$$

Apply the truncation $W_3(K) \rightarrow W_2(K)$, a ring homomorphism commuting with $\wp$ (Fact 1.2): $(0, 0, f)$ maps to 0, so $(s_0, s_1) \equiv (\beta_0, \beta_1) \pmod{\wp W_2(K)}$ with $\beta_0, \beta_1 \in F_0$. Thus the $\mathbb{Z}/4$ -torsor $(s_0, s_1)_K$ is the base change of a torsor over F_0 , and $\text{ed}_k((s_0, s_1)_K) \leq \text{trdeg}_k F_0 = 1$. But $\text{ed}_k((s_0, s_1)_K) = 2$ by Fact 1.4(b). $\square$

Corollary 5.2. *For our witness, $F_0 = \iota(k(B)) \not\subseteq K$.*

Proof. Lemma 2.6 provides $\beta \in W_3(F_0)$ with $s \equiv \beta$ over K' . $\square$

5.2. The twisted pencil.

Proposition 5.3 (pencil proposition). *Suppose there is $u \in k(C_2)$ with $[k(C_2) : k(u)] = 2$ such that*

- (H1) $g^*u = u + 1$ for the generator g of G_2 (hence $(g^i)^*u = u + i$);
- (H2) $D_0 \sim u^*(a)$ for some (equivalently every) $a \in \mathbb{P}^1(k)$.

Then $F_0 \subseteq K$, contradicting Corollary 5.2. In particular no such u exists.

Proof. (1) The quotient $B = \mathbb{P}_t^1$. Put $t := u^2 + u = q \circ u$ with $q: \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$, $w \mapsto w^2 + w$ (a morphism of degree 2 with $q(\infty) = \infty$). By (H1), t is G_2 -invariant, so $k(t) \subseteq k(C_2)^{G_2} = k(B)$; and $[k(C_2) : k(t)] = \deg(q \circ u) = \deg q \cdot \deg u = 4 = [k(C_2) : k(B)]$. Hence $k(B) = k(t)$: $B = \mathbb{P}_t^1$, and the quotient morphism $t: C_2 \rightarrow B$ is the composite $q \circ u$.

(2) The K -rational coordinate $\tilde{u}$. Put

$$\tilde{u} := u \otimes 1 + 1 \otimes z_0 \in k(C_2) \otimes_k L_2 \subseteq L_2(Y).$$

By (H1) and $g(z_0) = z_0 + 1$ (Section 1), $(g^* \otimes g)(\tilde{u}) = (u + 1) \otimes 1 + 1 \otimes (z_0 + 1) = \tilde{u}$; so $\tilde{u} \in L_2(Y)^{G_2} = K(E_2)$ (Lemma 2.9(b)). It defines a K -morphism $\tilde{u}: E_2 \rightarrow \mathbb{P}_K^1$ whose base change to L_2 is, via ψ , the morphism $Y \rightarrow \mathbb{P}_{L_2}^1$ given by $u + z_0$, that is $\tau_{z_0} \circ u_{L_2}$ with $\tau_{z_0}: \mathbb{P}_{L_2}^1 \rightarrow \mathbb{P}_{L_2}^1$, $w \mapsto w + z_0$. For $c \in \mathbb{P}^1(K)$ let $F_c := \tilde{u}^*(c)$, an effective K -rational divisor on E_2 (Fact 1.12(d)). By Fact 1.12(d),

$$\psi(F_c \otimes_K L_2) = (\tau_{z_0} \circ u_{L_2})^*(c) = u_{L_2}^*(\tau_{z_0}^{-1}c) = \begin{cases} u_{L_2}^*(c + z_0) & c \in K, \\ u_{L_2}^*(\infty) & c = \infty, \end{cases}$$

so $\deg F_c = \deg u = 2$ and $\deg \tilde{u} = 2$.

(3) *Descent of the linear equivalence to K .* Any two points of $\mathbb{P}^1(L_2)$ are linearly equivalent on $\mathbb{P}_{L_2}^1$, so all fibres $u_{L_2}^*(a')$, $a' \in \mathbb{P}^1(L_2)$, are linearly equivalent on Y ; and $u_{L_2}^*(a) = u^*(a) \otimes_k L_2$ for $a \in \mathbb{P}^1(k)$. Hence, by (H2) and Proposition 3.1(a),

$$\psi(F_c \otimes L_2) \sim u^*(a) \otimes L_2 \sim D_0 \otimes L_2 \sim \Delta = \psi(D_Q \otimes L_2) \quad \text{on } Y,$$

and transporting along the isomorphism ψ^{-1} : $D_Q \otimes_K L_2 \sim F_c \otimes_K L_2$ on $E_2 \otimes_K L_2$. By Fact 1.12(b), $D_Q \sim F_c$ on E_2 for every $c \in \mathbb{P}^1(K)$. (This is the second gap flagged by the referee: the linear equivalence obtained over L_2 does descend to K , because $\text{Pic}(E_2) \rightarrow \text{Pic}(E_2 \otimes L_2)$ is injective.)

(4) *The K -rational members of $|F_c|$ are the $\tilde{u}$ -fibres.* Take $c = 0$ and write $\Phi := \tilde{u}^*(0) = \text{div}_0(\tilde{u})$. By Fact 1.12(a), $\ell(\Phi) = \ell(\Phi \otimes L_2) = \ell(u^*(a) \otimes L_2) = \ell(D_0) = 2$. The functions 1 and $\tilde{u}^{-1}$ lie in $L(\Phi)$ (indeed $\text{div}(\tilde{u}^{-1}) + \Phi = \text{div}_\infty(\tilde{u}) \geq 0$) and are linearly independent over K since $\tilde{u}$ is not constant; so $L(\Phi) = K \cdot 1 \oplus K \cdot \tilde{u}^{-1}$. An effective divisor D' on E_2 with $D' \sim \Phi$ is $D' = \Phi + \text{div } h$ with $0 \neq h = \alpha + \beta \tilde{u}^{-1}$, $\alpha, \beta \in K$:

- if $\beta = 0$, $D' = \Phi = \tilde{u}^*(0)$;
- if $\alpha = 0$, $D' = \text{div}_0(\tilde{u}) - \text{div}(\tilde{u}) = \text{div}_\infty(\tilde{u}) = \tilde{u}^*(\infty)$;
- if $\alpha\beta \neq 0$, $h = \alpha(\tilde{u} - c')/\tilde{u}$ with $c' := \beta/\alpha \in K$, and $D' = \text{div}(\tilde{u} - c') - \text{div}(\tilde{u}) + \text{div}_0(\tilde{u}) = \text{div}(\tilde{u} - c') + \text{div}_\infty(\tilde{u}) = \text{div}_0(\tilde{u} - c') = \tilde{u}^*(c')$, using $\text{div}_\infty(\tilde{u} - c') = \text{div}_\infty(\tilde{u})$.

Since D_Q is effective and $D_Q \sim \Phi$ by (3), $D_Q = \tilde{u}^*(c')$ for some $c' \in \mathbb{P}^1(K)$.

(5) $\tilde{u}(Q) = c'$. The point Q : $\text{Spec } K' \rightarrow E_2$ factors through $\xi_Q \subseteq \text{Supp } D_Q = \text{Supp } \tilde{u}^*(c') = \tilde{u}^{-1}(c')$. Hence $\tilde{u} \circ Q$: $\text{Spec } K' \rightarrow \mathbb{P}_K^1$ factors through the closed point $c' \in \mathbb{P}_K^1$, whose residue field is K ; a K' -point of $\mathbb{P}_K^1$ with image a K -rational closed point c' is the point c' itself. So $\tilde{u}(Q) = c' \in \mathbb{P}^1(K) \subseteq \mathbb{P}^1(K')$.

(6) $t_E = q_K \circ \tilde{u}$. Let $q_K: \mathbb{P}_K^1 \rightarrow \mathbb{P}_K^1$ be $w \mapsto w^2 + w + s_0$, $\infty \mapsto \infty$. By (1) and Lemma 2.9(c), the base change of $t_E: E_2 \rightarrow B_K = \mathbb{P}_K^1$ to L_2 is, via ψ , $t \times \text{id} = q_{L_2} \circ u_{L_2}$; the base change of $q_K \circ \tilde{u}$ is $q_{L_2} \circ \tau_{z_0} \circ u_{L_2}$, and

$$q_{L_2}(\tau_{z_0}(w)) = (w + z_0)^2 + (w + z_0) + s_0 = w^2 + w + (z_0^2 + z_0 + s_0) = w^2 + w$$

because $z_0^2 + z_0 = s_0$ and the characteristic is 2. So the two morphisms agree after base change to L_2 , hence are equal (Fact 1.12(c)).

(7) *Conclusion.* By (5) and (6), $b = t_E(Q) = q_K(\tilde{u}(Q)) = q_K(c') \in \mathbb{P}^1(K)$. By Lemma 2.11, b has image the generic point of $B = \mathbb{P}_K^1$; if $c' = \infty$ then $b = \infty$ is a k -point, which is excluded. So $c' \in K$ and b is the K -point $c'^2 + c' + s_0$ of $\mathbb{P}_K^1$, i.e. $b^*(t) = c'^2 + c' + s_0 \in K$. By Lemma 2.11, $\iota(t) = b^*(t)$ and $F_0 = \iota(k(B)) = \iota(k(t)) = k(\iota(t)) \subseteq K$. $\square$

Remark 5.4. The formula $\tilde{u} = u + z_0$ is the case $n = 1$ of the coordinate $h = y - z$ of [P, Prop. 5.2(a)]: a K' -point of the twist of $\mathbb{P}_u^1$ (with G_2 acting through $u \mapsto u + 1$) is a G_2 -equivariant map $\text{Spec } L'_2 \rightarrow \mathbb{P}_u^1$, i.e. an element $u(Q)$ of $L'_2 \cup \{\infty\}$ with $g(u(Q)) = u(Q) + 1$; then $u(Q) + z_0$ is G_2 -invariant, i.e. lies in K' . Our argument works with the function $\tilde{u}$ on E_2 rather than with points, so that the linear system $|F_c|$ can be used.

5.3. Verification of the hypotheses in the hyperelliptic case.

Proposition 5.5 (Step 3). *$g(C_2) \geq 2$ is impossible.*

Proof. Assume $g(C_2) \geq 2$ and let $u: C_2 \rightarrow \mathbb{P}^1$ be the degree-2 map of Corollary 4.5, with $D_0 \sim u^*(a)$; write $L(D_0) = \langle h_0, h_1 \rangle$, $u = h_1/h_0$, so that $k(u) \subseteq k(C_2)$ is the subfield generated by the ratios of elements of $L(D_0)$ (these ratios are exactly the Möbius transforms $(\alpha + \beta u)/(\gamma + \delta u)$).

(i) G_2 preserves $k(u)$. Let $h \in \text{Aut}(C_2)$. Then h^*D_0 has degree 2 and $\ell(h^*D_0) = \ell(D_0) = 2$ (h^* is an automorphism of $k(C_2)$ mapping $L(D_0)$ onto $L(h^*D_0)$), so $h^*D_0 \sim D_0$ by Fact 1.13(d): $h^*D_0 = D_0 + \text{div } h'$ and $L(h^*D_0) = h'^{-1}L(D_0)$. A ratio of two elements of $L(D_0)$ is mapped

by h^* to a ratio of two elements of $L(h^*D_0)$, which is a ratio of two elements of $L(D_0)$. Hence $h^*(k(u)) = k(u)$, and $h \mapsto \rho(h) := h^*|_{k(u)}$ is a homomorphism $\text{Aut}(C_2) \rightarrow \text{Aut}_k(k(u)) = \text{PGL}_2(k)$ (the group G_2 is abelian, so the contravariance of $h \mapsto h^*$ is harmless).

(ii) $\ker \rho|_{G_2} = \{1, g^2\}$ and $\rho(g)$ has order 2. The kernel of ρ on $\text{Aut}(C_2)$ consists of the automorphisms acting trivially on $k(u)$, i.e. of the elements of $\text{Gal}(k(C_2)/k(u)) = \{1, \iota_u\}$ (Corollary 4.5). By Fact 1.8, $\rho(g)$ has order ≤ 2 , so $g^2 \in \ker \rho \cap G_2 \subseteq \{1, \iota_u\}$; as $g^2 \neq 1$, $g^2 = \iota_u$ and $\ker \rho \cap G_2 = \{1, g^2\}$; since $g \notin \ker \rho$, $\rho(g)$ has order exactly 2.

(iii) *Normal form.* By Fact 1.8, $\rho(g)$ is conjugate in $\text{PGL}_2(k)$ to $u \mapsto u + 1$; conjugating means replacing u by a Möbius transform $u' = (\alpha + \beta u)/(\gamma + \delta u)$, which is again a degree-2 function with $u'^*(a) \sim u^*(a') \sim D_0$ (the fibres of u' are the fibres of u). After this change of coordinate, $g^*u = u + 1$.

So (H1) and (H2) of Proposition 5.3 hold, and Proposition 5.3 gives a contradiction. $\square$

6. STEP 4: GENUS ONE

In this section $g(C_2) = 1$. We choose an origin and regard C_2 as an elliptic curve E ; $g \in \text{Aut}(E)$ is an automorphism of order 4 of the curve E .

6.1. Faithful $\mathbb{Z}/4$ -actions on elliptic curves in characteristic 2.

Lemma 6.1. *Let E be an elliptic curve over k with origin O and $\varphi \in \text{Aut}(E)$ an automorphism of order 4; write $\varphi = t_P \circ \alpha$ with $P \in E(k)$, $\alpha \in \text{Aut}(E, O)$ (Fact 1.14(a)).*

- (a) *If E is supersingular, then α has order 4 and there is $R \in E(k)$ with $t_R \circ \varphi \circ t_{-R} = \alpha$; in particular φ fixes the point $-R$.*
- (b) *If E is ordinary, then $\alpha = 1$ and $\varphi = t_P$ with P of order 4.*

Proof. The projection $\text{Aut}(E) \rightarrow \text{Aut}(E, O)$ is a homomorphism, so $\alpha^4 = 1$. If $\alpha = 1$, $\varphi = t_P$ has order $\text{ord } P$. If $\alpha^2 = 1 \neq \alpha$, then $(\alpha - 1)(\alpha + 1) = 0$ in the domain $\text{End}(E)$ gives $\alpha = -1$, and $\varphi^2 = t_P \alpha t_P \alpha = t_P t_{\alpha P} \alpha^2 = t_{P-P} = 1$, so φ does not have order 4. Hence either $\alpha = 1$ and $\text{ord } P = 4$, or α has order 4. (a) If E is supersingular, $E(k)[2] = 0$ (Fact 1.14(b)), so $E(k)$ has no element of order 4 and α has order 4. For $R \in E(k)$, $t_R \varphi t_{-R} = t_R t_P \alpha t_{-R} = t_{R+P} t_{-\alpha R} \alpha = t_{P+(1-\alpha)R} \alpha$ (Fact 1.14(a)). Since $\alpha \neq 1$, $1 - \alpha$ is a nonzero isogeny, surjective on $E(k)$ (Fact 1.14(c)); choose R with $(1 - \alpha)R = -P$. Then $t_R \varphi t_{-R} = \alpha$, and $\varphi(-R) = t_{-R} \alpha t_R(-R) = t_{-R}(\alpha(O)) = -R$. (b) If E is ordinary, $j(E) \neq 0$ and $\text{Aut}(E, O) = \{\pm 1\}$ (Fact 1.14(b)), so α has order ≤ 2 , hence $\alpha = 1$ and $\text{ord } P = 4$. $\square$

Remark 6.2. [A, Lemma C.2] (= [E, Lemma 1.3]) says more in the supersingular case: the six elements of order 4 of $\text{Aut}(E, O) \cong \text{SL}_2(\mathbb{F}_3)$ are conjugate, so (E, φ) is isomorphic to $(E_0, \text{Witt action})$ of [P, Theorem 6.2], with $E_0: y_1^2 + y_1 = y_0^3 + y_0^2$ and $(y_0, y_1) \mapsto (y_0 + 1, y_1 + y_0)$. We do not need this identification: the abstract argument below only uses the fixed point of (a).

6.2. The supersingular case.

Proposition 6.3 (Step 4(i)). *$g(C_2) = 1$ with C_2 supersingular is impossible.*

Proof. Let $E = C_2$. By Lemma 6.1(a) the automorphism g of E has a fixed point, which we call ∞ and take as the origin; then $g = \alpha \in \text{Aut}(E, \infty)$ is a group automorphism of order 4 (Fact 1.14(a)).

(a) $E(k)^\alpha = 0$. In the domain $\text{End}(E)$, $(\alpha^2 - 1)(\alpha^2 + 1) = \alpha^4 - 1 = 0$ and $\alpha^2 \neq 1$, so $\alpha^2 = -1$. With $a := \alpha + \hat{\alpha} \in \mathbb{Z}$ and $\deg \alpha = 1$, Fact 1.14(c) gives $\alpha^2 - a\alpha + 1 = 0$, i.e. $a\alpha = 0$, so $a = 0$ because $\text{End}(E)$ is torsion-free and $\alpha \neq 0$. Hence $\deg(1 - \alpha) = 1 + \deg \alpha - a = 2$, and $\ker(1 - \alpha)(k) = \{R \in E(k) : \alpha R = R\}$ is a group of order ≤ 2 (Fact 1.14(c)), hence killed by 2, hence contained in $E(k)[2] = 0$ (Fact 1.14(b)).

(b) $D_0 \sim 2\infty$. By Proposition 3.1(b), $g^*D_0 \sim D_0$, and $g^*\infty = \infty$ since g fixes ∞ ; so the class $[D_0 - 2\infty] \in \text{Pic}^0(E)$ is g^* -invariant. Under the group isomorphism $E(k) \rightarrow \text{Pic}^0(E)$, $R \mapsto [R - \infty]$ [Sil, III.3.4], g^* corresponds to $R \mapsto g^{-1}(R) = \alpha^{-1}(R)$ (the pullback of the point divisor R along the automorphism g is the point $g^{-1}(R)$, and $g^{-1}(\infty) = \infty$). Let $R \in E(k)$ with $[R - \infty] = [D_0 - 2\infty]$; then $\alpha^{-1}R = R$, so $R \in E(k)^\alpha = 0$ by (a), $R = \infty$, and $D_0 \sim 2\infty$.

(c) *A coordinate with $g^*u = u + 1$.* By Fact 1.13(a), $\ell(2\infty) = 2$ and $\ell(\infty) = 1$; let $L(2\infty) = \langle 1, u \rangle$. Then u has pole divisor exactly 2∞ , so $\deg u = [k(E) : k(u)] = 2$ and $u^*(\infty) = 2\infty$. Since $g(\infty) = \infty$, g^* maps $L(2\infty)$ into itself ($\text{div}(g^*h) = g^*\text{div} h \geq -2g^*\infty = -2\infty$), so $g^*u = au + b$ with $a \in k^\times$, $b \in k$. As $(g^*)^4 = 1$ on $L(2\infty)$, comparing the coefficient of u gives $a^4 = 1$, i.e. $(a + 1)^4 = 0$, $a = 1$. If $b = 0$ then u would be G_2 -invariant, i.e. $u \in k(B)$, whence $[k(E) : k(u)] \geq [k(E) : k(B)] = 4$, a contradiction. So $b \neq 0$, and replacing u by u/b we get $g^*u = u + 1$.

Now (H1) holds, and (H2) holds because $D_0 \sim 2\infty = u^*(\infty)$ by (b). Proposition 5.3 gives a contradiction. $\square$

Remark 6.4. In the model $(E_0, \text{Witt action})$ one can take $u = y_0$ and $t = y_0^2 + y_0$; this is the picture of [P, Theorem 6.2], and [A, Corollary 11.7] is the special case $C = \bar{U}_3$ (where $C_2 = E_0$ and $E_2 = E' \otimes_{K_2} K$). The argument above is the generalisation to an arbitrary $\mathbb{Z}/8$ -curve C with supersingular $C/\langle g^4 \rangle$; it needs neither [P, Prop. 7.5] nor the explicit equations.

6.3. The ordinary case: the corestriction argument.

Proposition 6.5 (Step 4(ii)). *$g(C_2) = 1$ with C_2 ordinary is impossible.*

Proof. Let $E = C_2$ with any origin O . By Lemma 6.1(b), $g = t_P$ with $P \in E(k)$ of order 4, so $G_2 = \{t_{iP}\}$ acts by translations. By Fact 1.14(d) there is a finite étale isogeny $\phi: E \rightarrow E'$ of degree 4 with kernel $\langle P \rangle$ and $E' \cong E/\langle t_P \rangle$; since $k(E/\langle t_P \rangle) = k(E)^{G_2} = k(B)$, we may take $B = E'$ (an elliptic curve) and $t = \phi$. We have the exact sequence

$$0 \rightarrow \mathbb{Z}/4 \xrightarrow{i \mapsto iP} E \xrightarrow{\phi} B \rightarrow 0$$

of Fact 1.15(c), with connecting maps $\delta_F: B(F) \rightarrow H^1(F, \mathbb{Z}/4)$.

(a) $\delta_{K'}(b) = \text{res}_{K'/K}(s_0, s_1)$, where $b = t_E(Q) \in B(K')$. By Lemma 2.9(c), $\phi \circ x = b \circ \text{pr}$ for $x = x_Q: \text{Spec } L'_2 \rightarrow E$. So x factors through a K' -morphism $x': \text{Spec } L'_2 \rightarrow \phi^{-1}(b) = E \times_{B,b} \text{Spec } K'$. The scheme $\phi^{-1}(b)$ is the $\mathbb{Z}/4$ -torsor of class $\delta_{K'}(b)$, with $1 \in \mathbb{Z}/4$ acting by $t_P = g_E$; the scheme $\text{Spec } L'_2$ is the $\mathbb{Z}/4$ -torsor of class $\text{res}_{K'/K}(s_0, s_1)$, with 1 acting by $\text{Spec}(g)$ (Lemma 2.3(b), Fact 1.2); and $x \circ \text{Spec}(g) = g_E \circ x$ (Lemma 2.7). Hence x' is a $\mathbb{Z}/4$ -equivariant morphism of $\mathbb{Z}/4$ -torsors over K' , so it is an isomorphism (Fact 1.15(b)), and the two classes coincide.

(b) $N_{K'/K}(b) \in B(k)$ and δ_K vanishes on it. $N(b) = b + \sigma b$ is σ -invariant, hence lies in $B(K)$ (Fact 1.10(a)), and $B(K) = B(k)$ by Fact 1.9(d) applied to the elliptic curve B over k and the purely transcendental extension $K = k(s_0, s_1, s_2)$ of k . For $b_0 \in B(k)$, the torsor $\phi^{-1}(b_0) \times_k \text{Spec } K$ is the base change of the $\mathbb{Z}/4$ -torsor $\phi^{-1}(b_0)$ over k , which has a k -point (ϕ is surjective on k -points) and is therefore trivial; so $\delta_K(b_0) = 0$.

(c) *Contradiction.* By Fact 1.15(c),(a) and (a),(b):

$$\begin{aligned} 0 &= \delta_K(N_{K'/K} b) = \text{cores}_{K'/K} \delta_{K'}(b) = \text{cores}_{K'/K} \text{res}_{K'/K}(s_0, s_1) \\ &= 2 \cdot (s_0, s_1) \quad \text{in } H^1(K, \mathbb{Z}/4) = W_2(K)/\wp. \end{aligned}$$

By Fact 1.1(d), $2 \cdot (s_0, s_1) = (0, s_0^2)$, and by Fact 1.1(b), $(0, s_0^2) - (0, s_0) = (0, s_0^2 + s_0) = \wp(0, s_0) \in \wp W_2(K)$; so $2 \cdot (s_0, s_1) \equiv (0, s_0)$, which is nonzero in $W_2(K)/\wp$ by Lemma 1.3. Contradiction. $\square$

Remark 6.6. (i) Step 4(ii) uses neither Step 1 nor the non-constancy of b : it shows that $\text{res}_{K'/K}(s_0, s_1)$ is not of the form $\delta_{K'}(b)$ for any $\mathbb{Z}/4$ -isogeny $E \rightarrow B$ of elliptic curves over k , any separable quadratic K'/K and any $b \in B(K')$. (ii) The identification $\mathbb{Z}/4 \cong \ker \phi$, $1 \mapsto P$, was chosen to match $g = t_P$; another choice would change δ by an automorphism of $\mathbb{Z}/4$ acting on H^1 , which does not affect the conclusion $2 \cdot \delta_{K'}(b) \neq 0$.

7. COMPLETENESS OF THE CASE ANALYSIS, PROOF OF THE THEOREM, COROLLARIES

Proposition 7.1 (completeness). *Let K' be a witness and $C_2 = C/\langle g^4 \rangle$ as in Section 2. Then exactly one of the following holds, and each leads to a contradiction:*

- (i) $g(C_2) = 0$: impossible by Lemma 4.1;
- (ii) $g(C_2) = 1$ and C_2 is supersingular: impossible by Proposition 6.3;
- (iii) $g(C_2) = 1$ and C_2 is ordinary: impossible by Proposition 6.5;
- (iv) $g(C_2) \geq 2$: impossible by Proposition 5.5.

Hence no witness exists.

Proof. The genus of C_2 is a non-negative integer, and an elliptic curve in characteristic 2 is either supersingular or ordinary; so the four cases are exhaustive and mutually exclusive. In cases (ii) and (iv) the input is the effective divisor D_0 of Proposition 3.1 with $\ell(D_0) = 2$ (Proposition 4.3); in case (iv) Corollary 4.5 and Proposition 5.5 apply, in case (ii) Proposition 6.3; case (iii) uses only Lemma 6.1(b) and the corestriction argument. Each case ends in a contradiction with one of: Fact 1.8 (case (i)); the dominance of x , i.e. the non-constancy of the descent point (Proposition 4.3, which is used in cases (ii) and (iv)); Corollary 5.2, i.e. Theorem 5.1 (cases (ii) and (iv), through Proposition 5.3); Lemma 1.3 (case (iii)). $\square$

Proof of Theorem 0.1. Let F'/K be a field extension of degree 2 and suppose $\text{ed}_k(\tau_{\text{gen}} \otimes F') \leq 1$. By Lemma 2.2, $F' = K(v_f)$ is a witness with $f \notin \{0, s_0\} + \wp K$, and Sections 2–6 apply; Proposition 7.1 gives a contradiction. Hence $\text{ed}_k(\tau_{\text{gen}} \otimes F') \geq 2$ for every quadratic F'/K . For $F' = K(v_2)$, Fact 1.4(f) gives $\text{ed}_k(\tau_{\text{gen}} \otimes K(v_2)) \leq 2$, so equality holds there, and $\text{ed}_k^{[2]}(\tau_{\text{gen}}) = \min_{[F':K] \leq 2} \text{ed}_k(\tau_{\text{gen}} \otimes F') = 2$ (the extension $F' = K$ contributes $\text{ed}_k(\tau_{\text{gen}}) \geq 2$ by Fact 1.4(f)). Finally $d_1(\tau_{\text{gen}}) = \min\{d : \text{ed}_k^{[d]}(\tau_{\text{gen}}) \leq 1\}$: since $\text{ed}^{[1]} = \text{ed}_k(\tau_{\text{gen}}) \geq 2$ and $\text{ed}^{[2]} = 2$, $d_1 \geq 3$; since $\text{ed}^{[4]} = 1$ (Fact 1.4(f)), $d_1 \leq 4$. $\square$

Corollary 7.2. (a) *For every quasi-projective generically free primitive G -variety Z over k of dimension ≤ 1 , the twist $\tau_{\text{gen}} Z$ has no free closed point of degree 1 or 2: $D(Z) \cap \{1, 2\} = \emptyset$ in the notation of [P, Definition 4.3]. In particular this holds for every curve Z with faithful G -action, and for $Z = \bar{U}_3$ it recovers [A, Corollary 11.7].*

- (b) $\text{ed}_k(\tau_{\text{gen}} \otimes K') \geq 2$ for every quadratic K'/K ; the families of [A, Theorem 11.9] and the extensions of [P, Theorem 7.3] are no longer special.
- (c) *The profile at $(2, 3)$ reads $\text{ed}^{[1]} \in \{2, 3\}$, $\text{ed}^{[2]} = 2$, $\text{ed}^{[3]} \in \{1, 2\}$, $\text{ed}^{[4]} = \dots = \text{ed}^{[7]} = 1$, $\text{ed}^{[8]} = 0$; $d_2 \in \{1, 2\}$, $d_1 \in \{3, 4\}$, $d_0 = 8$. Whether $\text{ed}^{[3]} = 1$ (a cubic witness) is untouched by this note: Steps 1 and 4(ii) use the norm over a quadratic extension and the σ -invariance of D_Q in an essential way.*

Proof. (a) A free closed point of degree $d \leq 2$ on $\tau_{\text{gen}} Z$ with residue field F' gives, by [P, Theorem 4.2, (2) $\Rightarrow$ (3) $\Rightarrow$ (1)], $[F' : K] = d$ and $\text{ed}_k(\tau_{\text{gen}} \otimes F') \leq 1$, contradicting Theorem 0.1. (b) is Theorem 0.1. (c) combines Theorem 0.1 with [P, Theorem 7.1] and [A, Theorem 11.16]. $\square$

7.1. The two “windows” of the external notes are closed. The addendum [A, Remark 11.12] and the note [S, Cor. 7.2] list two remaining shapes for a quadratic witness: (1) the descent class is the étale $\mathbb{Z}/8$ -isogeny torsor of an ordinary elliptic curve at a non-constant point

(“isogeny shape”), and (2) f a polynomial in s_2 of degree ≥ 8 (a witness curve which is the Artin–Schreier–Witt curve of a polynomial Witt vector). Both are excluded by Theorem 0.1, since no witness exists at all. For (1) the corestriction argument gives the exclusion directly and at level 3, and shows that the “necessary condition” of [S, Prop. 7.1] is vacuous:

Proposition 7.3 (the isogeny shape is impossible). *Let $0 \rightarrow \mathbb{Z}/8 \rightarrow C \xrightarrow{\phi} C' \rightarrow 0$ be an étale isogeny of elliptic curves over k with cyclic kernel of order 8, let K'/K be a separable quadratic extension, and let $b \in C'(K')$. Then $\delta_{K'}(b) \neq \text{res}_{K'/K}(s)$ in $H^1(K', \mathbb{Z}/8)$. In particular $\tau_{\text{gen}} \otimes K'$ is never of the isogeny shape, for any K' and any b (constant or not), and the group $C'(K)/\phi(C(K))$ of [S, Prop. 7.1] is $C'(k)/\phi(C(k)) = 0$.*

Proof. Suppose $\delta_{K'}(b) = \text{res}_{K'/K}(s)$. As in Proposition 6.5(b),(c), $N_{K'/K}(b) \in C'(K) = C'(k)$ (Fact 1.9(d)), δ_K vanishes on $C'(k)$, and Fact 1.15 gives $0 = \delta_K(Nb) = \text{cores } \delta_{K'}(b) = \text{cores res}(s) = 2s$. By Fact 1.1(d), $2s = (0, s_0^2, s_1^2)$, and

$$(0, s_0^2, s_1^2) - (0, s_0, s_1) = V(F(s_0, s_1) - (s_0, s_1)) = V\wp(s_0, s_1) = \wp V(s_0, s_1) \in \wp W_3(K)$$

(Fact 1.1(c)); so $2s \equiv (0, s_0, s_1) = V(s_0, s_1)$, which is nonzero in $W_3(K)/\wp$ by Lemma 1.3. Contradiction. The last assertion: $C'(K) = C'(k)$ and ϕ is surjective on k -points. $\square$

Remark 7.4. (i) Thus the assertion in [S, Prop. 7.1] that “ $Q + \sigma Q$ has exact order 8 in $C'(K)/\phi(C(K))$ ” cannot hold: its hypothesis is contradictory. (ii) For a witness whose cover $C \rightarrow B$ is étale with $g(B) \geq 2$ one could run the same norm argument after pulling the cyclic cover back from a $\mathbb{Z}/8$ -isogeny of $\text{Jac}(B)$ along an Abel–Jacobi map; this is not needed, since such a C has $g(C_2) \geq 2$ and falls under Step 3 (or Step 2).

8. WHAT IS CITED WITHOUT PROOF

For a reader checking the argument line by line, here is the complete list of external inputs; everything else in this note is proved above.

From [P].

- §2.2: the Artin–Schreier–Witt isomorphism $H^1(F, \mathbb{Z}/2^n) = W_n(F)/\wp$ and its compatibility with truncation, V , and base change (Fact 1.2); the formulas (2), (3), (4) (Fact 1.1(a)–(c)); (3) and its consequence (d) are re-derived, (5) is reproved as (e)).
- §2.3: L/K is Galois with group G , $L^{\langle g^{2^j} \rangle} = k(z_0, \dots, z_{j-1}, s_j, \dots)$, and $G/\langle g^4 \rangle$ acts on (z_0, z_1) by the Witt translation of W_2 with $\wp(z_0, z_1) = (s_0, s_1)$ (used for T_2 , Lemma 2.3).
- Lemma 2.1 ($\text{ed}_k(\tau_{\text{gen}}) = \text{ed}_k(\mathbb{Z}/8) \leq 3$); Fact 2.2(a) (twist points; we use it only to identify our E_2 with the twist, Construction 2.8); Lemma 2.3 and Theorem 4.2, (1) $\Rightarrow$ (3) and (2) $\Rightarrow$ (1) (Fact 1.6, Corollary 7.2(a)); Lemma 2.4 (Fact 1.7); Lemma 2.5(ii) (Fact 1.8, reproved); Proposition 3.2(c); Proposition 3.6; Proposition 3.9; Lemma 4.1(i); Lemma 4.1(ii) (Fact 1.9(d), whose proof is quoted); Proposition 6.1 ($\text{ed}_k(\mathbb{Z}/4) = 2$); Theorem 7.1 (upper bounds $\text{ed}^{[2]} \leq 2$ via $K(v_2)$, $\text{ed}^{[4]} = 1$, and $\text{ed}_k(\tau_{\text{gen}}) \geq 2$).

From [A]. Nothing is used without proof: Theorem 11.6(b) is reproved as Theorem 5.1, and Lemma C.2 is replaced by Lemma 6.1. Corollary 11.7, Proposition 11.8, Theorems 11.9–11.10 and Corollary 11.11 are only referred to in remarks.

Standard references.

- Galois descent for quasi-projective schemes, descent of morphisms and of points, $X(M)^\Gamma = X(F)$ [BLR, 6.1–6.2], [SeAG, V.20] (Fact 1.10).
- Existence and structure of the Picard scheme of a curve over k ; Pic^d is a torsor under the Jacobian; $\text{Pic}(C_F) = \text{Pic}_{C/k}(F)$ when $C(k) \neq \emptyset$ [BLR, 8.1, 8.2, 8.4], [Kl] (Fact 1.11).

- Weil’s extension theorem and the absence of rational curves on abelian varieties [Mi, Thm. 3.1, Cor. 3.8] (Fact 1.9(a),(b)).
- Flat base change for H^0 , morphisms to $\mathbb{P}^1$ and fibre divisors [Ha, II.6, III.9.3]; Riemann–Roch [Ha, IV.1.3] (Fact 1.12(a),(d), Fact 1.13(a)); arithmetic genus of a $(2, 2)$ -curve on $\mathbb{P}^1 \times \mathbb{P}^1$ and $g(\tilde{\Gamma}) \leq p_a(\Gamma)$ (proof of Fact 1.13(d)).
- Elliptic curves [Sil]: $\text{Aut}(E) = E(k) \rtimes \text{Aut}(E, O)$ (III.4.8); dual isogeny, trace and degree (III.6.2–6.3); $\# \ker \leq \deg$ (III.4.10); quotients by finite subgroups (III.4.12); supersingularity in characteristic 2 (V.3.1, Ex. 5.7); $\text{Aut}(E, O)$ for $j \neq 0$ and $j = 0$ (App. A, Prop. 1.2); $E(k) \cong \text{Pic}^0(E)$ (III.3.4) (Facts 1.14).
- Group cohomology: restriction, corestriction as morphisms of δ -functors, $\text{cores} \circ \text{res} = [F' : F]$, inflation–restriction [NSW, Ch. I, §1.5], [Se, I.2, I.5]; torsors and H^1 [Se, I.5] (Fact 1.15).
- Artin’s theorem on fixed fields; quotients of curves by finite groups; the smooth projective model of a curve and functoriality of its automorphisms.

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